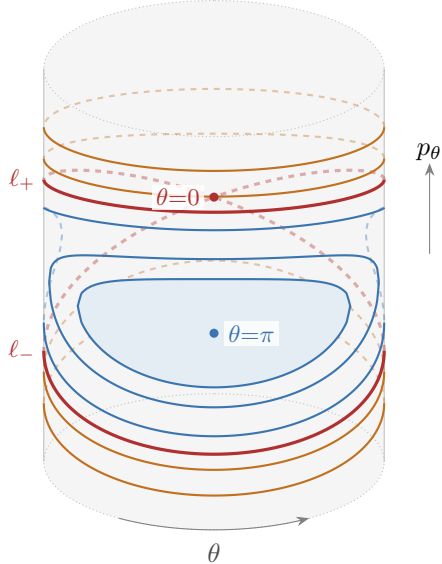


$\mathcal{M} \subset \mathbb{R}^4$: the solution manifold, the image of the cotangent lift. It is a cylinder $S^1 \times \mathbb{R}$, drawn here in $\mathbb{R}^3$ — see the caption.

librating, $H < 1$: a loop that *bounds* in $\mathcal{M}$. $\oint p_\theta d\theta$ is the symplectic area of the disk it encircles, so every symplectic chart must report the same number.



rotating, $H > 1$: a loop that runs *around* the cylinder. It bounds no surface in $\mathcal{M}$; its action uses the canonical one-form $p_\theta d\theta$, independently of the chart.

separatrix, $H = 1$: the two branches $\ell_\pm$, labelled by the sign of p_θ , each with action $+8$ along the flow. They meet at the unstable equilibrium $\theta = 0$.